CCIS 727 Assignment 7 — Analysis draft

INCOMPLETE EXECUTION: the cloud proxy returned 403 Forbidden for the official CIFAR-10 test download and the OpenAI CLIP checkpoint host. No predictions or accuracy results have been measured. This document does not claim successful execution. A successful notebook run will replace this draft with a report generated from actual measurements.

1. Alignment and frozen components

The course reading, “CLIP” (PDF pp. 1–2), describes separate image and text encoders projected into a shared embedding space. Contrastive pretraining brings matching image–text representations closer and separates mismatched examples. Comparing a normalized image vector with normalized class-description vectors therefore supports zero-shot classification by maximum cosine similarity. The shared 512-dimensional output permits these dot products. In this lab all pretrained image/text encoder and projection parameters are frozen, with evaluation mode and gradients disabled; no classifier is trained [1].

2. Prompt comparison and errors — pending execution

The prespecified comparison is class name alone (A) versus “a photo of a {class}.” (B), with official class names, identical preprocessing and image embeddings. The implementation selects ten official test images per class without replacement using NumPy default_rng(42), sorts indices within class, and uses batches of 16. A/B accuracy, per-class accuracy, B–A difference, all four paired correctness counts, confusion matrices and five-image findings are unavailable until execution succeeds. Neither prompt is asserted to be better. Visible-evidence explanations must await the actual selected images and predictions.

3. CLIP versus a generative VLM

CLIP ranks independently encoded candidate texts against an image; it does not generate an open-ended answer. As described in “VLM Architectures” (pp. 4–5) and “MLP Adapter” (p. 6), LLaVA combines a vision encoder with a visual projector/adapter and a language model. The projector maps visual features into the language model’s input representation; the language model conditions on these visual tokens and a question to generate text [2]. Running LLaVA is not part of this experiment.

4. Resolution and visual token compression

“Dynamic Resolution” (p. 4) explains that downscaling can lose image details; “Cross Attention Adapter” (p. 5) discusses compressing visual embeddings with learned queries and the trade-off between information and sequence length. Fine-detail tasks could suffer when relevant information is removed or compressed. This experiment cannot test that claim directly: its original images are all 32×32, the official preprocessing is fixed, and neither input resolution nor token compression is varied. Upsampling cannot recover lost source detail.

5. Limits and one controlled extension

A 100-image balanced sample permits only ten observations per class: one image changes total accuracy by one percentage point and class accuracy by ten. CIFAR-10’s low resolution can obscure object details and preserve background cues. Similar images may have occurred during CLIP pretraining; this lab has not established their absence. Results would measure transfer without task-specific training, not prove unseen evaluation content. One controlled extension would freeze the same pipeline and prompts and evaluate a prespecified larger, disjoint stratified test sample, without retuning, to assess sampling variability.

References

Course reading: Vision Language Models.pdf, dated August 14, 2025. Section/page citations above refer to the supplied PDF, not printed page numbers.

[1] Radford et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. ICML, PMLR 139. https://proceedings.mlr.press/v139/radford21a.html. Official CLIP source and zero-shot examples adapted: https://github.com/openai/CLIP, revision d05afc436d78f1c48dc0dbf8e5980a9d471f35f6.

[2] Liu et al. (2023). Visual Instruction Tuning. NeurIPS. https://arxiv.org/abs/2304.08485.

Dataset: Krizhevsky (2009), Learning Multiple Layers of Features from Tiny Images; official CIFAR-10 test split via torchvision.datasets.CIFAR10. https://www.cs.toronto.edu/~kriz/cifar.html.